

Theory of Average-Reward Markov Decision Processes

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Markov Decision Process

A finite **Markov Decision Process (MDP)** is a tuple $M = (\mathcal{S}, \mathcal{A}, P, R)$:

- **State-space $\mathcal{S}$** (with size S)
- **Action-space $\mathcal{A}$** (with size A)
- **Transition function P** : Selecting $a \in \mathcal{A}$ in $s \in \mathcal{S}$ leads to a transition to s' with probability $P(s'|s, a)$. $P(\cdot|s, a)$ is a probability distribution over $\mathcal{S}$, i.e.,

$$\sum_{s' \in \mathcal{S}} P(s'|s, a) = 1$$

- **Reward function R** : Selecting $a \in \mathcal{A}$ in $s \in \mathcal{S}$ yields a reward $r \sim R(s, a)$.
- The action-space may generally be state-dependent; if necessary, we use $\mathcal{A}_s$ to denote the set of actions available in state s .
- In general, $\mathcal{S}$ or $\mathcal{A}$ could be finite, countably infinite, or continuous.

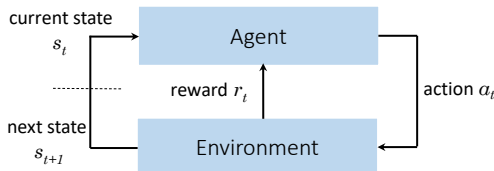


Interaction with MDP

An **agent** interacts with the MDP for N rounds.

At each time step t :

- The agent observes the current state s_t and takes an action $a_t \in \mathcal{A}$
- The environment (MDP) decides a reward $r_t := r(s_t, a_t) \sim R(s_t, a_t)$ and a next state $s_{t+1} \sim P(\cdot | s_t, a_t)$
- The agent receives r_t (any time in step t before start of $t + 1$)



This interaction produces a trajectory (or history)

$$h_t = (s_1, a_1, r_1, s_2, a_2, r_2, \dots, s_{t-1}, a_{t-1}, r_{t-1}, s_t)$$



Classification of MDPs based on N

- **Finite-Horizon MDPs:** $N < \infty$, and the goal is to solve

$$\max_{\text{all strategies}} \mathbb{E} \left[\sum_{t=1}^{N-1} r(s_t, a_t) + r(s_N) \right]$$

- **Infinite-Horizon Discounted MDPs:** $N = \infty$, and given **discount factor** $\gamma \in (0, 1)$, the goal is to solve

$$\max_{\text{all strategies}} \mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^{t-1} r(s_t, a_t) \right]$$

- **Infinite-Horizon Undiscounted MDPs (Average-Reward MDPs):** $N = \infty$, and the goal is to solve

$$\max_{\text{all strategies}} \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \left[\sum_{t=1}^N r(s_t, a_t) \right]$$

This lecture: We study Average-Reward MDPs.



The Optimality Criterion

Let's consider optimizing the **N -step cumulative reward** (a.k.a. **total reward**):

$$\sup_{\text{all strategies}} \mathbb{E} \left[\sum_{t=1}^N r_t \right]$$

$\Rightarrow$ An ill-defined objective as it could grow unbounded when $N \rightarrow \infty$, *even with bounded rewards*.

We instead consider maximizing the **average expected reward**:

$$\sup_{\text{all strategies}} \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \left[\sum_{t=1}^N r_t \right]$$

- Hence the name average-reward MDPs.
- A well-defined objective.
- It also makes sense in practice. (More on this later.)



Assumption on Rewards

We assume:

- Deterministic rewards so that

$$r_t = r(s_t, a_t) = R(s_t, a_t)$$

- Bounded rewards in the following sense:

$$|r(s, a)| \leq R_{\max} < \infty$$

Extension to stochastic reward is done by replacing $r(s, a)$ with the $\mathbb{E}[r(s, a)]$ in the results.



Policy

When interacting with an MDP, actions are taken according to some **policy**. Policies classes are defined identically as in discounted MDPs, where a policy may be:

- deterministic or randomized (stochastic)
- history-dependent or stationary

	Deterministic	Randomized
Stationary	$\pi : \mathcal{S} \rightarrow \mathcal{A}$	$\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$
History-dependent	$\pi : \mathcal{H} \rightarrow \mathcal{A}$	$\pi : \mathcal{H} \rightarrow \Delta(\mathcal{A})$

- $\Delta(\mathcal{A})$ denotes the simplex of probability distributions over $\mathcal{A}$.
- $\mathcal{H}$ the set of all possible history sequences (trajectories)
- For a randomized policy π , $\pi(a|s)$ denotes the probability of choosing a in s .



Policy

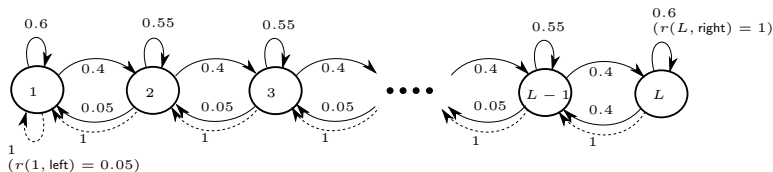
	Deterministic	Randomized
Stationary	$\pi : \mathcal{S} \rightarrow \mathcal{A}$	$\pi : \mathcal{S} \rightarrow \mathcal{A}$
History-dependent	$\pi : \mathcal{H} \rightarrow \mathcal{A}$	$\pi : \mathcal{H} \rightarrow \Delta(\mathcal{A})$

- Π^{SD} : Stationary deterministic policies
- Π^{SR} : Stationary randomized policies
- Π^{HD} : History-dependent deterministic policies
- Π^{HR} : History-dependent randomized policies

$$\begin{aligned} \text{(i)} \quad & \Pi^{\text{SD}} \subset \Pi^{\text{SR}} \subset \Pi^{\text{HR}} \\ \text{(ii)} \quad & \Pi^{\text{SD}} \subset \Pi^{\text{HD}} \subset \Pi^{\text{HR}} \end{aligned}$$



Example 1

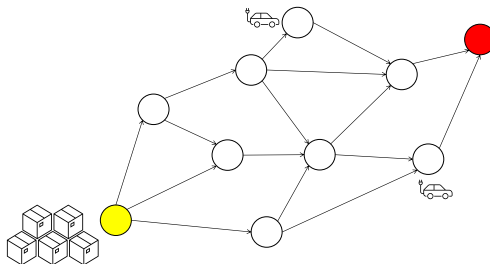


A continual task in RiverSwim

- **Variant 1:** The agent interacts with RiverSwim for an unspecified number N of round.
- **Variant 2:** If in state L and taking 'right', *Kystvagten* brings the agent to a random state, and the task repeats—the corresponding transition is not shown here.



Example 2



- **Task:** Transporting an arbitrary number of packages between source (in yellow) and destination (in red).
- The transportation cost differs across paths, and we are interested in minimizing the total cost.
- One package per round. Occasionally, a charging station must be visited.



Gain and Bias



Gain vs. Value

- For discounted and finite-horizon MDPs we defined notions of value function to distinguish the quality of various policies.
- Value functions measure the sum of future (discounted) rewards starting from any state.
- This machinery does not carry over to average-reward MDPs as cumulative reward could grow without bound.
- Instead, we define the notion of gain and bias to rank policies.



Gain

The **gain function** of policy π is a mapping $g^\pi : \mathcal{S} \rightarrow \mathbb{R}$ defined as

$$g^\pi(\textcolor{red}{s}) := \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E}^\pi \left[\sum_{t=1}^N r(s_t, a_t) \middle| s_1 = \textcolor{red}{s} \right].$$

where $\mathbb{E}^\pi$ indicates expectation over trajectories generated by π .

- $g^\pi(s)$ measure the per-step reward obtained under π starting from s , in the long run.
- The limit may not exist for all policies.
- For all π and s :

$$|g^\pi(s)| \leq R_{\max},$$

where $R_{\max}$ is an upper bound on the rewards.



Optimization using Gains

Solving an average-reward MDP M amounts to solving the following optimization problem:

$$g^*(s) = \sup_{\pi \in \Pi^{\text{HR}}} g^\pi(s),$$

for all $s \in \mathcal{S}$.

- $g^* : \mathcal{S} \rightarrow \mathbb{R}$ is called the **optimal gain**.
- Any policy achieving $g^*(s)$ for all s is called **gain-optimal** (or optimal, for short) and denoted by π^* .
- Do we have other optimality criteria? Discussion in class.



Is Gain Sufficient?

- Is gain alone is sufficient? $\implies$ Yes, if only the steady-state regime of MDP is concerned.
- However, for finite N ,

$$\mathbb{E}^{\pi} \left[\sum_{t=1}^N r(s_t, a_t) \middle| s_1 = s \right] \neq N g^{\pi}(s)$$

- The difference $\mathbb{E}^{\pi} \left[\sum_{t=1}^N r(s_t, a_t) \middle| s_1 = s \right] - N g^{\pi}(s)$ reflects the **transient rewards**.
- To capture the difference due to the transient regime we define **bias**.



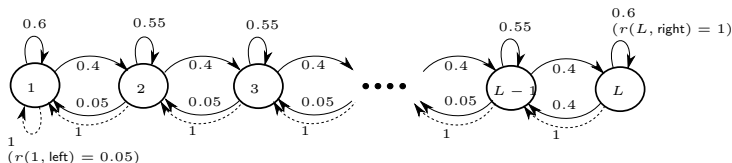
Bias

The **bias function** (or simply, bias) of policy π is a mapping $b^\pi : \mathcal{S} \rightarrow \mathbb{R}$ defined as

$$b^\pi(s) := \mathbb{E}^\pi \left[\sum_{t=1}^{\infty} \left(r(s_t, a_t) - g^\pi(s_1) \right) \middle| s_1 = s \right].$$

where $\mathbb{E}^\pi$ indicates expectation over trajectories generated by π .

- Assume $g^\pi(s) = g$ is constant, i.e., the MDP *forgets* the initial state—for example, it holds in RiverSwim for π prescribing to take 'right' action.
- Then $b^\pi(s) - b^\pi(s')$ indicates how much reward could have been obtained by starting in s rather than in s' .



Digression: Classification of MDPs Based on Reachability



MDP Classes

A classification of MDPs in terms of **reachability of various states**:

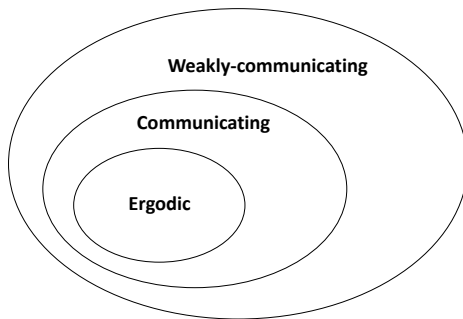
- ① An MDP is **ergodic** if it is possible to reach any state from any other state under every $\pi \in \Pi^{\text{SD}}$.
- ② An MDP is **communicating** if it is possible to reach any state from any other state under some $\pi \in \Pi^{\text{SD}}$.
- ③ An MDP is **weakly communicating** if its state-space can be partitioned into two sets:
 - (i) a set that is transient under every $\pi \in \Pi^{\text{SD}}$; and
 - (ii) a closed set in which every two states can reach each other under some $\pi \in \Pi^{\text{SD}}$.

In words, a weakly communicating MDP $\equiv$ a communicating MDP + some extra transient states.



MDP Classes

Hierarchy of MDP classes:



Diameter

Connectivity in MDPs can be measured via **diameter** (Jaksch et al., 2010).

Diameter of MDP

Let $T^\pi(s', s)$ denote the first hitting time of state s' when following $\pi \in \Pi^{\text{SD}}$ from $s (\neq s')$ in an MDP M . The diameter D of M is defined as

$$D := \max_{s \neq s'} \min_{\pi \in \Pi^{\text{SD}}} \mathbb{E}[T^\pi(s', s)].$$

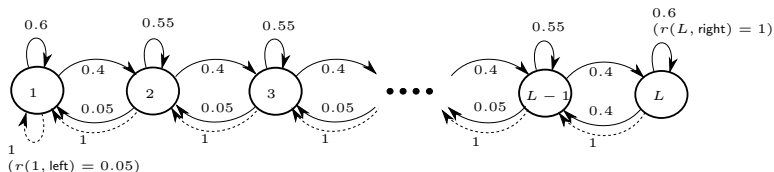
- Intuitively, D measures the worst-case shortest-path in the MDP:

$$D := \underbrace{\max_{s \neq s'}}_{\text{worst-case}} \underbrace{\min_{\pi \in \Pi^{\text{SD}}} \mathbb{E}[T^\pi(s', s)]}_{\text{shortest-path for } s \rightarrow s'}.$$

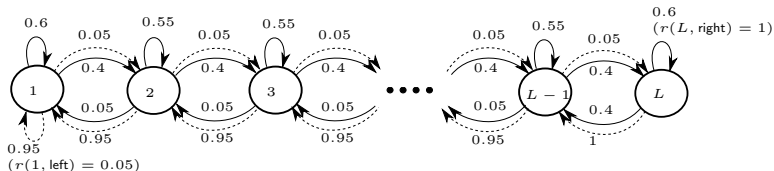
- MDP M is communicating $\iff M$ has a finite diameter.
- We may have $D = \infty$ for a weakly communicating MDP.



Example: RiverSwim



Is this MDP ergodic? Is it communicating?



Is this MDP ergodic? Is it communicating?



MDP Classes: Gain

- In **ergodic** MDPs, g^π for **any** π does not depend on the starting state, i.e., $g^\pi(s) = g^\pi$ for all s .
- In **weakly-communicating** MDPs, g^* does not depend on the starting state, i.e., $g^*(s) = g^*$ for all s .

	ergodic	communicating	weakly-communicating
g^π	constant	(possibly) state-dependent	(possibly) state-dependent
g^*	constant	constant	constant
D	finite	finite	(possibly) infinite

From now on, we only consider weakly-communicating MDPs.



Finding a Gain-Optimal Policy



Optimal Policy

In a weakly-communicating MDPs, at least one stationary deterministic policy exists, which is gain-optimal.

Hence,

$$g^*(s) = g^* = \sup_{\pi \in \Pi^{\text{HR}}} g^\pi(s) = \max_{\pi \in \Pi^{\text{SD}}} g^\pi(s)$$

- Hence, we can restrict attention to $\pi \in \Pi^{\text{SD}}$.
- Such optimal policy in $\pi \in \Pi^{\text{SD}}$ can be characterized using [Bellman optimality equations](#).



Bellman Optimality Equations

Theorem

If M is weakly communicating, then:

$$g^* + b^*(s) = \max_{a \in \mathcal{A}} \left(r(s, a) + \sum_{x \in \mathcal{S}} P(x|s, a) b^*(x) \right), \quad \forall s \in \mathcal{S}.$$

Furthermore, $\pi \in \Pi^{SD}$ is optimal if and only if:

$$\pi(s) \in \operatorname{argmax}_{a \in \mathcal{A}} \left(r(s, a) + \sum_{x \in \mathcal{S}} P(x|s, a) b^*(x) \right), \quad \forall s \in \mathcal{S}.$$

- $b^* : \mathcal{S} \rightarrow \mathbb{R}$ is called **the optimal bias function**.
- g^* is uniquely defined. (Why?)
- But b^* is defined up to an additive constant: If b^* is a solution, so is $b^* + c\mathbf{1}$ for any $c \in \mathbb{R}$.



VI

- We can use Value Iteration to solve Bellman Optimality Equations.
- The update is similar to the one in discounted MDPs:

$$V_{n+1}(s) = \max_{a \in \mathcal{A}} \left(r(s, a) + \sum_{x \in \mathcal{S}} P(x|s, a) V_n(x) \right), \quad s \in \mathcal{S}.$$

- V_n could grow unbounded. Yet we can show that $V_{n+1} - V_n$ could converge (to g^*).
- Hence, we choose to stop as soon as

$$\max_{s \in \mathcal{S}} (V_{n+1}(s) - V_n(s)) - \min_{s \in \mathcal{S}} (V_{n+1}(s) - V_n(s)) < \varepsilon$$

Or $\text{sp}(V_{n+1} - V_n) < \varepsilon$, where ‘sp’ denotes the **span operator** (or span semi-norm) defined as

$$\text{Given } f : \mathcal{S} \rightarrow \mathbb{R}^{\mathcal{S}}, \quad \text{sp}(f) := \max_{s \in \mathcal{S}} f(s) - \min_{s \in \mathcal{S}} f(s).$$



VI

- **input:** ε
- **initialization:** Select $V_0 \in \mathbb{R}^S$ arbitrarily. Set $n = 0$.
- **repeat:**
 - (ii) $n = n + 1$.
 - (i) Update, for each $s \in S$,

$$V_n(s) = \max_{a \in \mathcal{A}} \left(r(s, a) + \sum_{x \in S} P(x|s, a) V_{n-1}(x) \right)$$

until $\text{sp}(V_n - V_{n-1}) < \varepsilon$

- **output:**

$$\pi^{\text{VI}}(s) = \operatorname{argmax}_{a \in \mathcal{A}} \left(r(s, a) + \sum_{x \in S} P(x|s, a) V_n(x) \right), \quad s \in S$$



VI: Convergence

Theorem

In weakly communicating MDPs,

- For any $V_0 \in \mathbb{R}^S$, $(V_n)_{n \geq 0}$ generated by VI satisfies,

$$\lim_{n \rightarrow \infty} (V_{n+1}(s) - V_n(s)) = g^*, \quad \forall s \in S.$$

- VI converges after *finitely many iterations*. Furthermore, π^{VI} is ε -*optimal*: For all $s \in S$, $g^{\pi^{VI}}(s) \geq g^* - \varepsilon$.

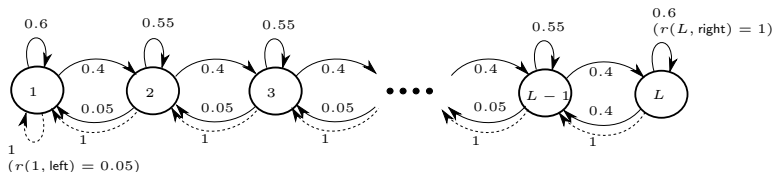
- $V_{n+1}(s) - V_n(s)$ for any s gives an approximation to g^* . It is best to approximate g^* as:

$$\frac{1}{2} \left[\max_{s \in S} (V_{n+1}(s) - V_n(s)) + \min_{s \in S} (V_{n+1}(s) - V_n(s)) \right]$$

- V_n also gives an approximation for b^* . So does $V_n - (\min_s V_n(s))\mathbf{1}$ (why?)



Example: RiverSwim



Optimal gain and optimal bias function in 6-state RiverSwim, computed via VI:

$$g^* = 0.467$$

$$b^*(1) = 0, \quad b^*(2) = 0.78, \quad b^*(3) = 2.04$$

$$b^*(4) = 3.37, \quad b^*(5) = 4.70, \quad b^*(6) = 6.03$$



Total Reward and Gain

N -step total reward, $\sum_{t=1}^N r_t$ is naturally connected to the average-reward.

Theorem

In weakly communicating MDPs, under π^ ,*

$$(i) \quad \mathbb{E} \left[\sum_{t=1}^N r_t \middle| s_1 = s \right] = Ng^* + \mathcal{O}(\text{sp}(b^*))$$

$$(ii) \quad \sum_{t=1}^N r_t \leq Ng^* + \mathcal{O}(\text{sp}(b^*)\sqrt{N \log(N/\delta)}), \quad w.p. \geq 1 - \delta$$

(i) is evident from the definition of bias function, and (ii) follows from Hoeffding's inequality.



Evaluating Gain and Bias

(outside of the scope of OReL)



Induced MRPs

Every $\pi \in \Pi^{\text{SR}}$ induces a **Markov reward process (MRP)** —defined identically as in discounted MDPs.

- The transition matrix P^π of MRP:

$$P_{s,s'}^\pi = \sum_{a \in \mathcal{A}} \pi(a|s) P(s'|s, a), \quad s, s' \in \mathcal{S}$$

- The reward vector of r^π of MRP:

$$r^\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) r(s, a), \quad s \in \mathcal{S}$$



Gain of Stationary Policies

Theorem

Let $\pi \in \Pi^{\text{SR}}$. Then, $g^\pi = \overline{P}^\pi r^\pi$, where

$$\overline{P}^\pi := \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N (P^\pi)^{t-1}$$

is the *limiting matrix* or the *Cesaro-average* of P^π ,

Proof. For $N \in \mathbb{N}$, the N -step accumulated reward in the MRP induced by π is

$$\begin{aligned} \mathbb{E}^\pi \left[\sum_{t=1}^N r(s_t, a_t) \middle| s_1 = s \right] \\ &= \mathbb{E}^\pi [r(s_1, a_1) | s_1 = s] + \mathbb{E}^\pi [r(s_2, a_2) | s_1 = s] + \dots + \mathbb{E}^\pi [r(s_N, a_N) | s_1 = s] \\ &= r^\pi(s) + [P^\pi r^\pi](s) + \dots + [(P^\pi)^{N-1} r^\pi](s) = \sum_{t=1}^N [(P^\pi)^{t-1} r^\pi](s) \end{aligned}$$

where we used that for any $t \geq 1$, when following $\pi \in \Pi^{\text{SR}}$,

$$\mathbb{P}(s_t = y | s_1 = x) = (P^\pi)_{x,y}^t.$$



Bellman Equation for Π^{SR}

Theorem (Bellman Equation for Policy π)

Let $\pi \in \Pi^{\text{SR}}$. Assume that π induces an MRP, which is irreducible or unichain. Then

$$g^\pi \mathbf{1} = \overline{P}^\pi r^\pi$$

Furthermore, the bias function b^π satisfies the Bellman equation:

$$g^\pi \mathbf{1} + (I - P^\pi)b^\pi = r^\pi.$$

As a result,

$$b^\pi = (I - P^\pi + \overline{P}^\pi)^{-1}(I - \overline{P}^\pi)r^\pi + c\mathbf{1},$$

where c is any arbitrary scalar.

Note that the matrix $I - P^\pi + \overline{P}^\pi$ is non-singular, so the last assertion above is well-defined.

